

**MTH622** Vectors and Classical Mechanics**Question No : 39 of 41**

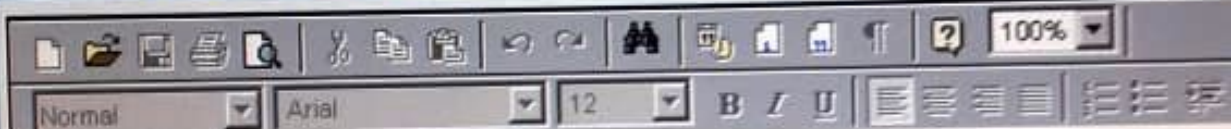
Find the moments of inertia of a ring of radius $2a$ about an axis through center.

Solution:

Let M be the mass and a the radius of the ring. Then the mass per unit length will be $M/2\pi 2a$

We regard this ring to be composed of small elements of mass (δm) each of length δs , we can write it as

$$\frac{\delta m}{\delta s} = \frac{M}{2\pi 2a} \Rightarrow \delta m = \frac{M}{2\pi 2a} \delta s$$

Answer (Please [click here](#) to Add Answer)

Moment of inertia of the element about an axis through center O and the perpendicular to the plane of the ring equals $\frac{M}{2\pi 2a} \delta s 4a^2$

Therefore the M.I of the whole ring will be

$$I_{ring} = \frac{M}{2\pi 2a} 4a^2 \sum_{elements} \delta s = \frac{Ma}{\pi} \int ds = \frac{Ma}{\pi} \times 2\pi 2a = 4Ma^2$$

Hence $4Ma^2$ is the required moment of inertia of the ring.

subjective solved by
khalil sahir

Start Time: 10:56 AM

39

108:00

Time Left



Done

Start

MathType (Lite mode) - ...

VU Examination System

Address <http://server/vutes/client/Paper.aspx>

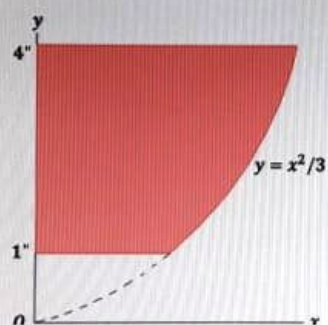
MTH622 Vectors and Classical Mechanics

MC18

Question No : 41 of 41

Marks: 5 (Budgeted Time 10 Min)

Calculate the moment of inertia of the shaded area given in figure about y-axis.



Solution

Here, we have

$$y = x^2/3 \quad (1)$$

We consider an elementary strip from the shaded area whose M.I is

$$I_{\text{strip}} = \frac{1}{3} x^2 (x dy) = \frac{1}{3} x^3 dy$$

Answer ([Please click here to Add Answer](#))

Then by integration, we obtain the moment of inertia of the whole shaded area

$$I_y = \sqrt{3} \int_1^4 y^{3/2} dy = \sqrt{3} \frac{2}{5} \left| y^{5/2} \right|_1^4$$

$$= \sqrt{3} \frac{2}{5} \left[(4)^{5/2} - (1)^{5/2} \right]$$

$$= \frac{2\sqrt{3}}{5} \left[(4)^{5/2} - (1)^{5/2} \right]$$

$$= \frac{2\sqrt{3}}{5} (32 - 1)$$

$$= 21.5 \text{ inch}$$

Address <http://server/vutes/client/Paper.aspx>

MTH622 Vectors and Classical Mechanics

MC180

Question No : 40 of 41

Marks: 5 (Budgeted Time 10 Min)

Calculate the moment of inertia of a uniform rod of length L about an axis perpendicular to the rod and passing through an end point.

(Let the x -axis be chosen along the length of the rod, with origin at one end point. Let M and a be the mass and length of rod respectively and suppose the rod to be composed of small elements.)

Let dm and dx be the mass and the length of the specific element of the rod at a distance x from the end point O .

Answer (Please [click here to Add Answer](#))

Then $\frac{dm}{dx} = \frac{M}{a}$

$$\Rightarrow dm = \frac{M}{a} dx$$

Then the moment of inertia of this element about the given axis is

$$I_{\text{element}} = dm x^2 = \frac{M}{a} x^2 dx$$

Hence the moment of inertia of the whole rod will be

$$\begin{aligned} I &= \sum_{\text{all elements}} \frac{M}{a} x^2 dx \\ &= \frac{M}{a} \int_0^a x^2 dx \\ &= \frac{M}{a} \cdot \left[\frac{x^3}{3} \right]_0^a = \frac{M}{a} \frac{a^3}{3} \\ &= \frac{1}{3} M a^2 \end{aligned}$$

Start Time: 10:56 AM

105:00

Time Left



subjective solved by
khalil sahir



Address: <http://server/vutes/client/Paper.aspx>

MTH622 Vectors and Classical Mechanics MC180403

Question No : 36 of 41 Marks : 3 (Budgeted Time : 5 Min)

Consider a small element of a uniform elliptic plate has moment of inertia $I = \delta m y^2$. Write down the expression for moment of inertia of whole plate.
(Note: Do not solve the expression)

Answer (Please click here to Add Answer)

Then the moment of inertia of whole plate will be

$$I_x = \int (\delta m) y^2 = \int_{plate} \rho ds y^2 = \frac{1}{4} M b^2$$

Start Time: 10:56 AM
106:00
Time Left

36

Done

Start MathType (Lite mode) - ... VU Examination System

Address: <http://server/vutes/client/Paper.aspx>

MTH622 Vectors and Classical Mechanics **MC1804**

Question No: 37 of 41 Marks: 3 (Budgeted Time 6 Min)

If a coordinate system rotates with angular velocity $\vec{\omega} = 2\hat{i} - 3\hat{j} + 5\hat{k}$ relative to the fixed coordinate system, and $\vec{A} = \sin t \hat{i} - \cos t \hat{j} + e^{-t} \hat{k}$ then find $d\vec{A}/dt$ with respect to the fixed point system.

Where $\left(\frac{d\vec{A}}{dt}\right)_r = -\sin t \hat{i} + \cos t \hat{j} - 2e^{-2t} \hat{k}$ and $\vec{\omega} \times \vec{A} = (4e^{-2t} - 3 \cos t)\hat{i} - (-2e^{-t} + 3 \sin t)\hat{j} + (\cos t + 2 \sin t)\hat{k}$

Answer (Please [click here](#) to Add Answer)

Normal Arial 12 B I U

$$\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r + \vec{\omega} \times \vec{A}$$

$$\left(\frac{d\vec{A}}{dt}\right)_f = \cos t \hat{i} + \sin t \hat{j} - e^{-t} \hat{k} + (-3e^{-t} + 5 \cos t)\hat{i} - (2e^{-t} - 5 \sin t)\hat{j} + (-2 \cos t + 3 \sin t)\hat{k}$$

$$= (\cos t + 5 \cos t - 3e^{-t})\hat{i} + (5 \sin t + \sin t - 2e^{-t})\hat{j} + (-e^{-t} - 2 \cos t + 3 \sin t)\hat{k}$$

$$= (6 \cos t - 3e^{-t})\hat{i} + (6 \sin t - 2e^{-t})\hat{j} + (-e^{-t} - 2 \cos t + 3 \sin t)\hat{k}$$

subjective solved by
khalil sahir

What is the impulse developed by a mass of 100 gm which changes its velocity from $i+j+k$ to $2i+2j+3k$ m/sec?

Answer (Please [click here](#) to Add Answer)

Solution:

Since we have the following information:

$$m = 100 \text{ gm} \quad v_1 = i + j + k \quad v_2 = 2i + 2j + 3k$$

As we know that

$$\text{Impulse} = p_2 - p_1 = mv_2 - mv_1 = m(v_2 - v_1)$$

$$\text{Impulse} = 100(2i + 2j + 3k - (i + j + k)) = 100(i + j + 2k)$$

Start Time: 10:56 AM

105:00

Time Left



38



Address: <http://server/vutes/client/Paper.aspx>

MTH622 Vectors and Classical Mechanics **MC180403**

Question No : 32 of 41 Marks: 2 (Budgeted Time 4 Min)

To find the inertia matrix of uniform solid cuboid (parallelepiped) at one of its corner, if $I_{xx} = \frac{M}{3}(b^2 + c^2)$

Then find I_{yy} , and I_{zz} .

Answer ([Please click here to Add Answer](#))

Normal Arial 12 B I U

$$I_{yy} = \frac{M}{3}(a^2 + c^2)$$
$$I_{zz} = \frac{M}{3}(a^2 + b^2)$$

File Edit View Tools Help

Address: <http://server/vutes/client/Paper.aspx>

MTH622 Vectors and Classical Mechanics **MC180**

Question No : 33 of 41 Marks: 2 (Budgeted Time 4 Min)

If moment of inertia of a uniform circular cylinder of length b and radius a about x-axis through the center and perpendicular to the central axis is:

$$I_x = m \left(\frac{1}{4} a^2 + \frac{1}{12} b^2 \right)$$

Answer ([Please click here to Add Answer](#))

Normal Arial 12 B I U

Due to symmetry we have

$$I_x = I_y = m \left(\frac{1}{4} a^2 + \frac{1}{12} b^2 \right)$$

Address <http://server/vutes/client/Paper.aspx>

MTH622 Vectors and Classical Mechanics

MC

Question No : 34 of 41

Marks: 2 (Budgeted Time 4 Min)

Write the expression for total energy of a particle executing Simple Harmonic Motion.

Answer ([Please click here to Add Answer](#))

Normal Arial 12 B / U

$$E = T + V$$

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

Start Time: 10:56 AM

106:00
Time Left



34



Done

Address <http://server/vutes/client/Paper.aspx>

MTH622 Vectors and Classical Mechanics

MC1804

Question No : 35 of 41

Marks: 3 (Budgeted Time 5 Min)

Two objects push off against one another starting from a stationary position. First object with mass 65 kg acquires a velocity of 2 m/s to the right. What velocity does the second object having mass 70 kg acquire?

Answer ([Please click here to Add Answer](#))

Rich text editor toolbar with icons for Bold, Italic, Underline, Text Color, Background Color, Bulleted List, Numbered List, Indent, Outdent, Link, Unlink, and others. The font is set to Arial, size 12.

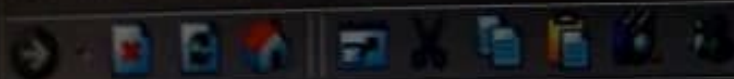
Start Time: 10:56 AM

106:00

Time Left

35

Navigation icons: Information, Previous, Next.



Address: <http://server/vubet/client/Paper.aspx>

MTH622 Vectors and Classical Mechanics

Question No : 28 of 41

If there is no net force acting on body, then its acceleration is

Answer (Please select your correct option)

zero

☒

constant

☐

increasing

☐

decreasing

☐

Start Time: 10:56 AM

109:00

Time Left



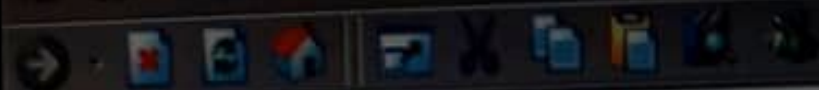
28

Done

Start

MathType (Lite mode)

VU Examination System

Address <http://server/vutes/client/Paper.aspx>**MTH622** Vectors and Classical Mechanics

Question No : 29 of 41

Total amount of mass and energy together in a system is

Answer (Please select your correct option)

☐ increasing☐ decreasing☐ zero☒ constant

Start Time: 10:56 AM

109:00

Time Left



29

Done

Start

MathType (Lite mode) - ...

VU Examination Syste...

Address <http://server/vites/client/Paper.aspx>

MTH622 Vectors and Classical Mechanics

Question No : 30 of 41

Measure of average kinetic energy of molecules is

Answer (Please select your correct option)

☒ temperature

☐ energy

☐ internal energy

☐ enthalpy

Start Time: 10:56 AM

108:00

Time Left



Done

Start

MathType (Lite mode) - ...

VU Examination System...

VU Examination System (CLIENT) VOTES 7 ... Spring 2019 (Final Term) ... - Microsoft

File Edit View Tools Help

Address <http://server/votes/client/Paper.aspx>

MTH622 Vectors and Classical Mechanics

Question No : 31 of 41

What is a degrees of freedom of the system of rigid bodies.

Answer ([Please click here to Add Answer](#))

Normal Arial 12 B I U

A rigid body has six degrees of freedom.

Start Time: 10:56 AM

107:00 Time Left

31

Done

Start MathType (Lite mode) - ... VU Examination Syste...

MTH622 Vectors and Classical Mechanics

Question No : 27 of 41

Until a force acts on a body, it's velocity is

Answer (Please select your correct option)

☐ zero

☒ constant

☐ increasing

☐ decreasing

Start Time: 10:56 AM

109:00

Time Left



Done

Start

MathType (Lite mode) - ...

VU Examination System

Address <http://server/vutes/client/Paper.aspx>

MTH622 Vectors and Classical Mechanics

Question No : 26 of 41

The time taken by the particle to complete one oscillation is called

Answer (Please select your correct option)

☐ amplitude

☒ time period

☐ frequency

☐ none of these

Start Time: 10:56 AM

109:00

Time Left



26

Done

Start

MathType (Lite mode) - ...

VU Examination System

MTH622 Vectors and Classical Mechanics

Question No : 25 of 41

The path of projectile is called its

Answer (Please select your correct option)

☐ velocity

☐ acceleration

☒ trajectory

☐ none of these

✓ VR

Start Time: 10:56 AM

109:00

Time Left



Done

Start

MathType (Lite mode) - ...

VU Examination System

In a conservative field, the total energy of particle _____ throughout the motion.

Answer (Please select your correct option)

☐ varies rapidly

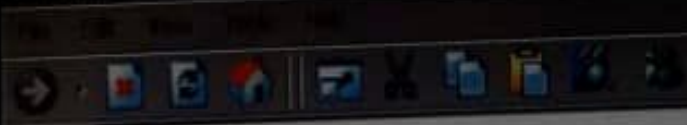
☐ does not remain constant

☒ remains constant

☐ none of these

✓ R

Start Time: 10:56 AM



Address <http://server/vutes/client/Paper.aspx>

MTH622 Vectors and Classical Mechanics

Question No : 20 of 41

Work done is said to be zero, when

Answer (Please select your correct option)

- ☐ a. Some force acts on a body, but displacement is zero
- ☐ b. No force acts on a body but some displacement takes place
- ☐ c. Either a) or b)
- ☐ d. None of these

Start Time: 10:56 AM

110:00
Time Left



20



MTH622 Vectors and Classical Mechanics

Question No : 21 of 41

The rate of change of displacement of a body is called

Answer (Please select your correct option)

☒ Velocity

☐ Acceleration

☐ Momentum

☐ None of these

Start Time: 10:56 AM

110:00

Time Left



21

Done

Start

MathType (Lite mode) - ...

VU Examination Syste...

File Edit View Tools Help

Address: <http://server/vutes/client/Paper.aspx>

MTH622 Vectors and Classical Mechanics

Question No : 22 of 41

For an irrotational flow

Answer (Please select your correct option)

☒ $\nabla \times \vec{v} = \vec{0}$

☐ $\nabla \times \vec{v} \neq \vec{0}$

☐ $\nabla \cdot \vec{v} = \vec{0}$

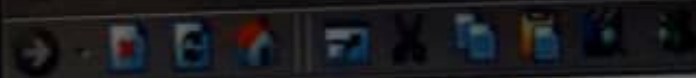
☐ None of these

Start Time: 10:56 AM

110:00
Time Left

Done

Start MathType (Lite mode) - ... VU Examination System...

**MTH622** Vectors and Classical Mechanics

Question No : 23 of 41

The path of a projectile in a non-resistive medium is

Answer (Please select your correct option)

☐ linear☐ Circular☒ Parabolic☐ Hyperbolic

Start Time: 10:56 AM

110:00

Time Left



23

Done

Start

MathType (Lite mode) - ...

YU Examination Syste...

Question No : 19 of 41

Marks: 1 (Budgeted Time 1 Min)

The periodic time of a particle with simple harmonic motion is _____ proportional to the angular velocity

Answer (Please select your correct option)

☐ Directly

☒ Inversely

☐ Square root

☐ None of these

Start Time: 10:56 AM

110:00

Time Left



19



Question No : 18 of 41

Marks: 1 (Budgeted Time 1 Min)

For a conservative holonomic dynamical system, the Lagrangian L , kinetic energy T and potential energy V are connected by

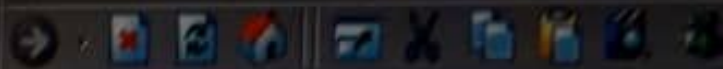
Answer (Please select your correct option)

☒ $L = T + V$

☐ $L = T - V$

☐ $L = 2T + V$

☐ $L = 2T - V$

Address <http://server/vutes/client/Paper.aspx>**MTH622** Vectors and Classical Mechanics

Question No : 17 of 41

At pivot point, body with mass ' m ' will have a torque by the

Answer (Please select your correct option)

☒ force☐☐ circle☐☐ central point☐☐ radius of circle

Start Time: 10:56 AM

112:00

Time Left



17

Done

Start

MathType (Lite mode) - ...

YU Examination System

According to parallel axis theorem, the moment of inertia of a section about an axis parallel to the axis through center of gravity (i.e. IP) is given by _____
(where, A = Area of the section, IG = Moment of inertia of the section about an axis passing through its C.G., and h = Distance between C.G. and the parallel axis.)

Answer (Please select your correct option)

☒ $IP = IG + Ah^2$

☐ $IP = IG - Ah^2$

☐ $IP = IG / Ah^2$

☐ $IP = Ah^2 / IG$

Start Time: 10:56 AM

113:00

Time Left



16



MTH622 Vectors and Classical Mechanics

Question No : 13 of 41

Cylinder has moment of inertia given by formula of

Answer (Please select your correct option)

☒ (ml^2)

☐

☐ mr

☐

☐ $\frac{1}{2}(mr)$

☐

☐ $5(mr^2)$

☐

Start Time: 10:56 AM

114:00

Time Left



Done

Start

MathType (Lite mode) - ...

VU Examination System

MTH622 Vectors and Classical Mechanics

Question No : 14 of 41

Moment of inertia of sphere is given as

Answer (Please select your correct option)

☐ $2(ml^2)$

☐ r^2

☐ $1/2(r)$

☒ $2/5(mr^2)$

✓ R

Start Time: 10:56 AM

114:00

Time Left



14

Done

Start

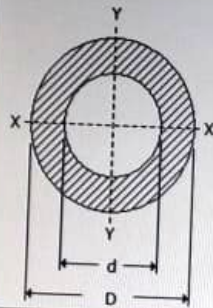
MathType (Lite mode) - ...

VU Examination System

Question No : 15 of 41

Marks: 1 (Budgeted Time 1 Min)

Moment of inertia of a hollow circular section, as shown in the below figure about X-axis, is



Answer (Please select your correct option)

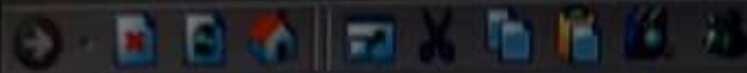
☐ $\pi/16 (D^2 - d^2)$

☐ $\pi/16 (D^3 - d^3)$

☐ $\pi/16 (D^4 - d^4)$

☒ $\pi/64 (D^4 - d^4)$

Done

Address <http://server/votes/client/Paper.aspx>**MTH622** Vectors and Classical Mechanics

Question No : 12 of 41

Moment of inertia of thin rod is given by formula of

Answer (Please select your correct option)

☒ $\frac{1}{12}(ml^2)$

☐

☐ mr^2

☐

☐ $2(mr)$

☐

☐ $\frac{2}{5}(mr^2)$

☐

Start Time: 10:56 AM

114:00

Time Left



12

Done

Start

MathType (Lite mode) - ...

VU Examination System

Address <http://server/vutes/client/Paper.aspx>

MTH622 Vectors and Classical Mechanics

MC180

Question No : 11 of 41

Marks: 1 (Budgeted Time 1 Min)

The moment of inertia of a circular section of base 'b' and height 'h' about an axis passing through its vertex and parallel to base is

Answer (Please select your correct option)

☐ $\frac{bh^3}{12}$

☐ $\frac{bh^3}{36}$

☐ $\frac{bh^3}{3}$

☒ $\frac{bh^3}{4}$



MTH622 Vectors and Classical Mechanics

Question No : 10 of 41

The wheels of a moving car possess

Answer (Please select your correct option)

☐ Potential energy only

☐ Kinetic energy of translation only

☐ Kinetic energy of rotation only

☒ Kinetic energy of translation and rotation both

Start Time: 10:56 AM

115:00

Time Left

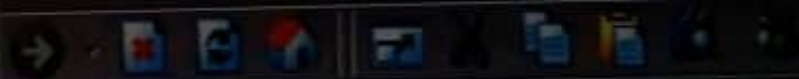


Done

Start

MathType (Lite mode) - ...

VU Examination System

Address <http://server/vutes/client/Paper.aspx>**MTH622** Vectors and Classical Mechanics

Question No : 9 of 41

The polar moment of inertia of a circular section is about

Answer (Please select your correct option)

☐ X-X axis☐ Y-Y axis☒ Z-Z axis☐ Neutral axis

Start Time: 10:56 AM

116:00

Time Left



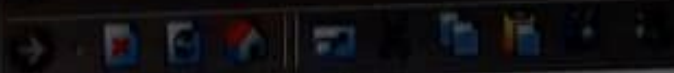
9

Done

Start

MathType (Lite mode) - ...

VU Examination System



Address: <http://server/vutes/client/Paper.aspx>

MTH622 Vectors and Classical Mechanics

Question No : 5 of 41

The functions in Green's theorem will be

Answer (Please select your correct option)

☐ Continuous derivatives

X

☐ Discrete derivatives

☐ Continuous partial derivatives

✓ R

☐ Discrete partial derivatives

Start Time: 10:56 AM

117:00

Time Left



5

Done

Start

MathType (Lite mode) - ...

VU Examination Syste...

Address <http://server/vutes/client/Paper.aspx>

MTH622 Vectors and Classical Mechanics

MC18

Question No : 6 of 41

Marks: 1 (Budgeted Time 1 Min)

If a time dependent vector function $\vec{A}$ is represented by $\vec{A}_f$ and $\vec{A}_r$ in fixed and rotating coordinate system, then

Answer (Please select your correct option)

☐ $\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r - \omega \times \vec{A}_r$

☒ $\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r + \omega \times \vec{A}_r$

☐ $\left(\frac{d\vec{A}}{dt}\right)_f = \vec{A}_r - \omega \times \vec{A}_r$

☐ $\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r - \vec{A} \times \vec{A}_r$

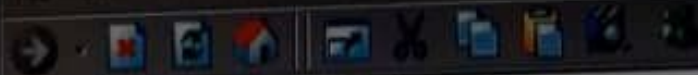
Start Time: 10:56 AM

116:00



6



**MTH622** Vectors and Classical Mechanics

Question No : 7 of 41

The axes relative to which product of inertia are zero are called

Answer (Please select your correct option)

- ☒ principal axes
- ☐ principal moment of inertia
- ☐ moment of inertia
- ☐ axes of rotation

Start Time: 10:56 AM

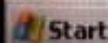
116:00

Time Left



7

Done



Start



MathType (Lite mode) - ...



VU Examination Syste...

A 3×3 symmetrical matrix has ----- real eigenvalues, which may be distinct or repeated

Answer (Please select your correct option)

☐ Infinite

☐ Zero

☐ Two

☒ Three

Start Time: 10:56 AM

116:00

Time Left



MTH622 Vectors and Classical Mechanics

Question No : 4 of 41

Mark

The Stoke's theorem can be used to find which of the following?

Answer (Please select your correct option)

☒ Area enclosed by a function in the given region

☐ Volume enclosed by a function in the given region

☐ Linear distance

☐ Curl of the function

Start Time: 10:56 AM

117:00

Time Left



Done

Start

MathType (Lite mode) - ...

VU Examination Syste...

Address: <http://server/votes/client/Paper.aspx>

MTH622 Vectors and Classical Mechanics

Question No : 3 of 41

$$\nabla \times \nabla \varphi = ?$$

Answer (Please select your correct option)

☐ $\nabla^2 \varphi$

☐ φ

☒ 0

☐ None of these

Start Time: 10:56 AM

117:00

Time Left



Done

Start

MathType (Lite mode) - ...

VU Examination System

Which of the following property is not true?

Answer (Please select your correct option)

☐ $\nabla \cdot (\vec{A} + \vec{B}) = \nabla \cdot \vec{A} + \nabla \cdot \vec{B}$

☐ $\nabla \cdot (\varphi \vec{A}) = \varphi (\nabla \cdot \vec{A}) + \vec{A} \cdot (\nabla \varphi)$

☐ $\nabla \cdot \nabla \varphi = \nabla^2 \varphi$

☒ $\nabla \cdot \vec{V} = \vec{V} \cdot \nabla$



R

The maximum value of the directional derivative of $\varphi(x, y, z)$ is equal to the

Answer (Please select your correct option)

☐ Magnitude of $\nabla \times \varphi$

☐ Magnitude of φ

☒ Magnitude of $\nabla \varphi$

☐ None of these

✓ R